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Seasonal differences and downstream transport of terrestrial vertebrate eDNA: Insights from integrated camera-trap survey and water eDNA metabarcoding

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ABSTRACT

Effective biodiversity conservation requires efficient, comprehensive, and cost-effective monitoring tools. While environmental DNA (eDNA) metabarcoding is widely used for aquatic biodiversity assessments, its application for monitoring terrestrial biodiversity remains relatively underexplored. In particular, the effects of seasonal variation on eDNA sampling efficacy and the downstream transport distances of terrestrial vertebrate eDNA in river systems are not well understood. We conducted a year-long survey (April 2023–April 2024) that combined camera trapping and eDNA sampling in the southwestern mountains of China, a global biodiversity hotspot. We collected water eDNA samples during two distinct periods representing different precipitation levels: October–November (low-rainfall season) and April (high-rainfall season). We compared seasonal effectiveness, evaluated costs, and developed an index to assess the relative importance of eDNA sampling sites. We also analyzed the elevational distributions of species detected by both methods and evaluated pairwise β -diversity patterns in relation to geographic distance to estimate the downstream transport distances. Our survey successfully detected 99 terrestrial vertebrate species through the combined efforts of camera trapping and eDNA sampling. This included 64 mammal, 27 bird, 1 reptile, and 7 amphibian species. Across both seasons, eDNA and camera trapping exhibited high complementarity in species detection, with 16 species detected by both methods. eDNA proved particularly cost-effective, detecting a more broad spectrum of taxa from ground-dwelling to arboreal animals compared to camera trapping survey. The relative importance of sampling sites varied between the low- and high-rainfall seasons, with mainstream and first-order tributary sites showing significantly higher importance in the low-rainfall season. Compared to camera trapping survey, water eDNA metabarcoding is more likely to detect the same species at lower-altitude sampling sites, reflecting upstream transport. Terrestrial vertebrate DNA was estimated to be transported downstream over distances of approximately 10–15 km. These findings provide valuable insights into sampling design and

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result interpretation in terrestrial vertebrate eDNA studies, underscoring the potential of eDNA metabarcoding as a reliable tool for monitoring terrestrial vertebrate biodiversity.

1. Introduction

Accelerating biodiversity loss is increasingly threatening ecosystems and the essential services they provide (Johnson et al., 2024). Despite some localized conservation successes, global biodiversity—particularly among terrestrial vertebrates—continues to decline at an alarming rate, with no clear indications of a slowdown (Ceballos et al., 2017; Dirzo et al., 2014; Barnosky et al., 2011; Butchart et al., 2010). Currently, nearly 7400 terrestrial vertebrate species are classified as threatened (i.e., Vulnerable, Endangered, or Critically Endangered) according to the IUCN Red List (IUCN, 2024), and since 1900, 198 vertebrate extinctions have been documented, a rate at least 10–100 times higher than background levels (Ceballos et al., 2015). Terrestrial vertebrates are disproportionately concentrated in mountainous regions, which cover about 25 % of the earth's land surface but harbor over 85 % of vertebrate species (Rahbek et al., 2019). The island-like properties of mountains, combined with habitat fragmentation, climate change, and other anthropogenic pressures, further escalate extinction risks for these species (Zhang et al., 2020). The loss or decline of vertebrate species can trigger cascading effects throughout ecosystems, disrupting interspecific interactions, altering community structures, and diminishing vital ecosystem functions such as seed dispersal (Ceballos et al., 2020; Bello et al., 2015; Tilman et al., 2014). These declines may also facilitate the spread of pathogens, exacerbating ecological imbalances (Ceballos et al., 2020). In light of these pressing threats, there is a critical need for comprehensive, rapid, and reliable biodiversity monitoring to guide conservation policies, inform timely interventions, and mitigate the loss of essential ecosystem services (Johnson et al., 2024; Stephenson, 2020; Proença et al., 2017; Tittensor et al., 2014).

Monitoring terrestrial vertebrate biodiversity can be achieved through a variety of traditional methods, generally categorized into capture-based approaches and observation-based surveys. Capture-based methods, such as pitfall traps, hand capturing, or the use of specialized traps, are designed to target specific taxa (Sutherland et al., 2006). In contrast, observation-based surveys include direct observations, call surveys for vocal species, and trace or fecal surveys, among others (Mena et al., 2021; Sutherland et al., 2006). While these traditional methods are widely used, they are often time-consuming, invasive, and taxonomically limited, depending heavily on the expertise of researchers (Deiner et al., 2017; Harrington et al., 2010; Stem et al., 2005). In recent years, remote monitoring technologies, such as satellite imagery, airborne sensors, and field-deployed devices (e.g., camera traps, sound recorders, and drones), have emerged to address some of these limitations (Buxton et al., 2018; Turner, 2014). These technologies enable monitoring across broader temporal and spatial scales. For instance, camera trapping offers a less invasive, long-term monitoring method, particularly useful in studies of terrestrial mammal diversity and ecology (Mena et al., 2021; Sales et al., 2020a). However, camera trapping remains costly, requires frequent maintenance, and is typically more effective for detecting larger species, with reduced efficacy for smaller animals (Burton et al., 2015). Despite these technological advancements, comprehensive and effective monitoring of terrestrial vertebrates remains challenging. The development of novel monitoring tools, combined with the integration of multiple complementary methods, is essential for generating a more comprehensive understanding of biodiversity and informing the development of targeted, effective conservation strategies (Ruppert et al., 2019; Turner, 2014).

Environmental DNA (eDNA)—mixture of genomic DNA shed by a variety of organisms in environmental samples (Ficetola et al., 2008)—can be derived from water (Li et al., 2023; Reji Chacko et al., 2023), soil (Leempoel et al., 2020), air (Lynggaard et al., 2024), or even leaf swabs (Lynggaard et al., 2023). Through metabarcoding, eDNA offers a non-invasive and rapid method for assessing biodiversity across multiple taxa. This approach has been widely applied to monitor aquatic organisms, such as fish and invertebrates, in both freshwater and marine environments (Girard et al., 2024; Perry et al., 2024; Lu et al., 2023). For terrestrial vertebrates, most eDNA studies have focused on evaluating its effectiveness for species detection, often by comparing it with other monitoring methods (Holm et al., 2023; Mena et al., 2021; Lyet et al., 2021; Leempoel et al., 2020; Shaw et al., 2016). While much of this research has concentrated on species presence or absence, eDNA has demonstrated significant potential for predicting species distribution patterns (Broadhurst et al., 2021). Furthermore, as a source of "novel community data" (Hartig et al., 2024), eDNA holds promise for uncovering the factors driving community assembly (Reji Chacko et al., 2023; Cantera et al., 2022).

As an important type of eDNA, water-derived eDNA has long been recognized for its ability to effectively monitor aquatic organisms, but also providing valuable insights into terrestrial species (Deiner et al., 2016; Sales et al., 2020b). In mountainous regions, where terrain complexity and limited accessibility pose challenges, water systems can accumulate eDNA from upstream catchments. This allows for efficient biodiversity monitoring over large spatial areas with minimal sampling effort, ensuring comprehensive representation of local biodiversity (Cristescu and Hebert, 2018).

Despite its potential for monitoring terrestrial vertebrates, eDNA metabarcoding faces challenges. One such challenge is the seasonal distribution of terrestrial vertebrates, particularly in subtropical regions (Hu et al., 2025). Seasonal activity of species is known to influence eDNA detection (De Souza et al., 2016), however, most studies have focused on a single season, overlooking the impacts of seasonal variation (Reji Chacko et al., 2023; Mena et al., 2021; Buxton et al., 2017). Research suggests that catchment-based sampling, such as sampling at the confluence of larger creeks, may be more cost-effective for capturing terrestrial mammal diversity compared to spatially distributed sampling (Lyet et al., 2021). However, the influence of seasonal variation and the relative contribution of tributaries in river networks on the effectiveness of such strategies remains unclear. There is also considerable uncertainty concerning the impacts of biotic and environmental conditions on the ecology of eDNA (its production, state, transport, and half-life within an environment; Barnes and Turner, 2016).

Some studies have investigated the transport of fish DNA in rivers through field experiments and modeling (Pont, 2024; Jane et al., 2015). However, research on the transport of terrestrial vertebrate eDNA in rivers remains limited. Deiner et al. (2016) and Sales et al. (2021) reported that riverine eDNA typically originates from both aquatic and terrestrial species throughout the entire upstream watershed. They proposed that biodiversity information from upstream regions converges downstream, with the river mouth acting as an “eDNA reservoir.” In contrast, Reji Chacko et al. (2023) observed significant β -diversity turnover among eDNA samples collected from low-, mid-, and high-altitude zones in the Swiss Alps, suggesting that biodiversity information in alpine rivers does not necessarily accumulate downstream. Furthermore, these studies have not been able to effectively quantify the transport distance of eDNA in river systems. This gap in knowledge complicates the evaluation of eDNA sampling effectiveness and the interpretation of resulting data, hindering reliable inference of species distribution models, and limiting the ability to make informed conservation decisions and implement targeted actions.

In this study, we compared the effectiveness of eDNA and camera trapping for monitoring terrestrial vertebrates across two seasons, while also evaluating the representativeness of sampling sites at different tributary levels. We adopted a systematic, seasonal, and hierarchical sampling design across representative areas in the mountains of southwestern China, a global biodiversity hotspot. Utilizing fine-scale occurrence data of terrestrial vertebrates, we also investigate β -diversity patterns in relation to geographic distance among sampling sites. The study addresses three central questions: (1) How do eDNA and camera trapping differ in terms of sampling costs and species composition when monitoring terrestrial vertebrates across seasons? (2) Is it necessary to sample across different tributary classes for eDNA collection in different seasons? (3) How far from the sampling point is the diversity of terrestrial vertebrates reflected by eDNA?

We hypothesized that: (1) eDNA sampling will be more cost-effective than camera trapping, but with both methods providing complementary species detection rates, consistent across seasons; (2) eDNA sampling in mountainous regions will capture relatively localized biodiversity, necessitating sampling across multiple tributary classes in both seasons to ensure comprehensive coverage; and (3) species diversity at downstream sampling sites will primarily reflect the species present in upstream areas. This study offers critical insights for effective biodiversity surveying and monitoring in remote mountain forests characterized by complex terrain and exceptionally high biodiversity.

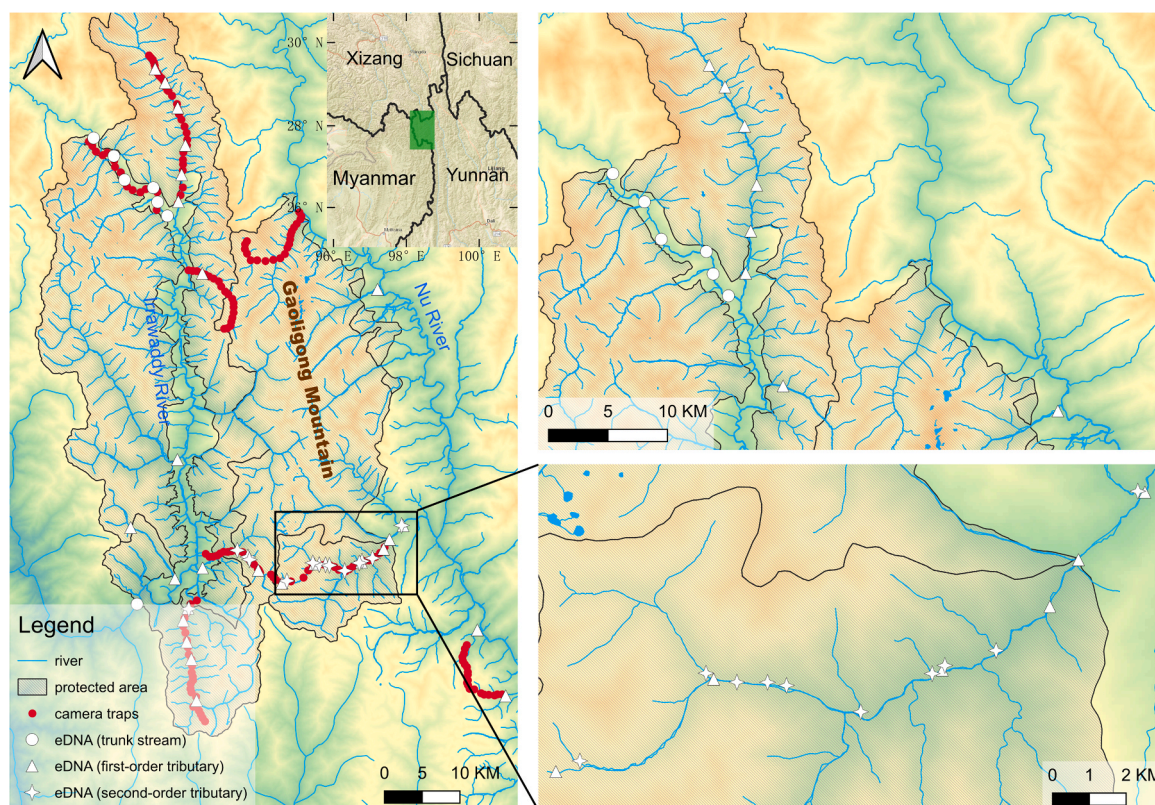


Fig. 1. The study area in the northern Gaoligong Mountains. Water eDNA sampling sites are distributed across the trunk streams, first-order tributaries, and second-order tributaries of the Dulong River and Nu River. Camera traps were primarily deployed along first-order tributaries in areas with minimal human disturbance.

2. Methods

2.1. Study area

The study was conducted in the northern region of the Gaoligong Mountains (GLGM) in Yunnan Province, China (98°08'–98°56'E, 27°29'–28°23'N), spanning an elevation range from 1170 to 5128 m and an area of 4381.53 km². Approximately 94.26 % of the region falls within designated protected zones, with forests accounting for 83.39 % of the total land cover (Zhou et al., 2023). The area is home to around 38,400 residents, predominantly concentrated in the valleys along the main channels of the Nu (Salween) River and the Dulong (Irrawaddy) River. Over 40 tributaries originate directly from the protected mountainous areas (Fig. 1).

The region is characterized by a subtropical monsoon climate, with an average annual temperature of 16°C and annual precipitation range of 2700 to 4700 mm, and a distinct alpine climatic pattern with both temperature and precipitation decreasing with increasing altitude. A defining climatic feature is the occurrence of a "double rainy season": the first from February to April, followed by a drier period in May, and the second rainy season from June to October (Zhou et al., 2023). The vegetation has clear altitudinal zonation, transitioning from evergreen broad-leaved forests at lower elevations to coniferous-broadleaved mixed forests, bamboo forests, and finally alpine scrub and meadows at higher elevations. Recent surveys report that the GLGM harbors rich biodiversity, including 212 mammal species (Li et al., 2024), 796 bird species (Wu et al., 2024), 107 reptile species, and 84 amphibian species (Wang et al., 2024). The region is also a critical habitat for numerous rare and endangered wildlife species.

2.2. Camera trapping

To conduct camera trapping, we selected eight sampling lines within the study area based on elevation range, forest area, habitat heterogeneity, and accessibility. Camera traps were deployed from April 2023 to April 2024. Along each sampling line, 20–30 camera traps were deployed, ensuring a minimum spacing of 500 m between individual traps, resulting in a total of 175 cameras. The camera traps were positioned 0.5–1 m above the ground, primarily along animal trails and away from human pathways to minimize disturbance. No bait or lures were used to attract animals. To reduce false triggers, the camera trap sensitivity was set to "low." Once activated, each camera trap was configured to capture two consecutive photos per trigger, with a 3-second interval between triggers. Batteries and SD cards were manually replaced approximately every six months during the deployment period. Captured images were manually reviewed, and species were identified to the lowest possible taxonomic level based on morphological characteristics and movement patterns. Independent events were defined as all images of the same species captured by the same camera trap within a single day (Li et al., 2018). We also recorded the effective working days for each camera trap. For each species, we estimated the naïve occupancy (NO) as the proportion of sites that recorded at least one independent event of the target species (Holm et al., 2023).

2.3. eDNA sampling

For water eDNA sampling, we established 47 sampling sites, including 7 in the trunk stream, 26 in ten first-order tributaries, and 14 in second-order tributaries. At each site, three replicate samples were collected. Each replicate consisted of 3 L of water, which was directly collected in the field using a 200 mL sterile syringe. The water was immediately filtered through a 6 cm diameter, 0.8 µm pore size PES disk filter and preserved in Longmire's solution. To prevent cross-contamination, a new 200 mL sterile syringe was used for each replicate. Negative controls were included to monitor potential contamination, with one negative control prepared for samples collected from the same first-order tributary (sampling line) or from different tributaries on the same day. Negative controls were generated by filtering pure water through a 200 mL sterile syringe and a 3.3 cm diameter PES filter (0.8 µm pore size). These negative controls were handled, transported, and stored alongside the field samples.

After sampling, we measured and recorded the geographic coordinates, river width, river depth, flow rate, water temperature, pH, dissolved oxygen, and electrical conductivity (EC) at each sampling site. Rainfall data for the sampling day and the day prior to sampling were obtained from the European Centre for Medium-Range Weather Forecasts (<https://www.ecmwf.int/>). All samples were stored and transported at room temperature and later preserved at –20°C in the laboratory. Sampling was conducted during October–November 2023 and April 2024. Based on observed precipitation patterns (Table S1) and the region's characteristic "double rainy season" phenomenon, we considered October–November 2023 as the low-rainfall season and April 2024 as the high-rainfall season.

2.4. Laboratory procedures

All laboratory work was performed in designated, separate facilities dedicated exclusively to DNA extraction and environmental DNA (eDNA) metabarcoding procedures. Both laboratories were equipped with positive air pressure systems, ultraviolet (UV) sterilization, and regular air renewal to minimize the risk of contamination. Stringent contamination control protocols were adhered to throughout the study, following the guidelines by Valentini et al. (2016). DNA extraction was carried out using the DNeasy Blood & Tissue Kit (Qiagen GmbH, Hilden, Germany), with a modified protocol based on Spens et al. (2016). Negative controls were processed first, followed by eDNA samples, with the final DNA elution volume set at 100 µL. Extracted DNA was stored at 4°C. In addition to the eDNA samples, a negative extraction control was included to detect potential cross-contamination during the extraction procedure.

Polymerase chain reaction (PCR) amplification was performed using the MT-RNR1 primer set (MT-RNR1_F: 5'-ACTGGGATTAGATACCCC-3', MT-RNR1_R: 5'-YRGACAGGCTCCTCTAG-3'; Ji et al., 2022; Riaz et al., 2011). This primer pair targets the mitochondrial 12S rRNA gene, which has been shown to exhibit high amplification efficiency across a broad spectrum of terrestrial

vertebrate taxa (Ji et al., 2022). Each sampling replicate was subjected to three independent PCR amplifications, and each PCR replicate was labeled with a unique combination of dual-index tags. This approach served two purposes: first, to reduce false-negative results caused by the stochastic nature of eDNA amplification, and second, to detect and eliminate “tag jumping” errors (Schnell et al., 2015) by applying the DAME protocol (Zepeda-Mendoza et al., 2016). In addition, a negative control was included in each PCR batch to monitor potential cross-contamination. PCR products that passed quality control were submitted to Novogene (Beijing, China) for PCR-free library preparation. Libraries were then subjected to 150 bp paired-end sequencing on the Illumina NovaSeq 6000 platform. All negative controls, including sampling, extraction, and PCR, were sequenced alongside the eDNA samples to monitor and detect potential DNA contamination.

2.5. Bioinformatics processing and taxonomic assignment

The bioinformatics workflow was primarily based on the methods described by Ji et al. (2022). Initial preprocessing of the raw sequencing data was performed using fastp, followed by the removal of sequences shorter than 100 bp with VSEARCH v2.28.1 (Rognes et al., 2016). Next, the DAME tool (Zepeda-Mendoza et al., 2016) was employed to identify and eliminate “tag jumping” and erroneous reads. Filtering criteria included a minimum sequence length of 75 bp, presence in at least two PCR replicates, and a minimum of four copies per unique sequence. All negative controls were also processed using DAME to detect potential contamination sources, allowing us to fine-tune the filtering parameters to remove contaminant sequences. Any residual chimeric sequences were eliminated with VSEARCH v2.28.1. The remaining sequences were clustered into preliminary operational taxonomic units (pre-OTUs) using the sumacust algorithm in OBITools (Boyer et al., 2016). The pre-OTUs exhibiting high sequence similarity and co-occurrence patterns were subsequently merged using the R package LULU (Frøslev et al., 2017) to generate the final OTUs and representative sequences.

For taxonomic assignment, we utilized PROTAX (Somervuo et al., 2017, 2016) on representative sequences of each OTU. PROTAX employs a library of local reference sequences, statistical multinomial regression models, and phylogenetic trees to estimate the probability that a given sequence belongs to each taxonomic rank. It also provides probabilities for sequences that may be classified as “unknown” (i.e., not present in the reference database). The PROTAX results were combined with BLAST search results to assign each OTU to the most likely taxonomic unit. OTUs identified as Human (*Homo sapiens*) were treated as contaminants and removed from all subsequent analyses. To ensure taxonomic accuracy, we cross-checked and updated the scientific names of species in accordance with the latest taxonomic revisions and surveys of terrestrial vertebrates in the Gaoligong Mountains (Li et al., 2024; Wang et al., 2024; Wu et al., 2024). For example, although 12S rRNA eDNA metabarcoding cannot reliably differentiate between *Naemorhedus evansi* (Burmese goral) and *Naemorhedus griseus* (Chinese goral), recent studies suggest that the gorals in this region are more likely to be *Naemorhedus evansi* (Li, 2020). The conservation status for identified species was compiled, including their designation on the China Key Protected Wild Animals List and their status in China’s Red List.

2.6. Statistical analysis

All statistical analyses and visualizations were conducted using R version 4.3.3 (R Core Team, 2024) and Microsoft Excel. To facilitate the analysis, OTUs corresponding to the same species, as identified by PROTAX, were merged into a single entry. The merged OTU table, along with the list of independent events from camera trapping, was transformed into a binary (0/1) presence-absence table for each species. For comparison with the eDNA results, camera trapping data recorded between April 30, 2023, and April 30, 2024, were extracted for subsequent analyses. Unless otherwise specified, terrestrial vertebrates in the following analyses refer exclusively to wild species.

We used the R package iNext (Hsieh et al., 2016) to plot species accumulation curves for both camera trapping and eDNA methods. Each method generated four separate curves representing all terrestrial vertebrates, mammals, birds, and amphibians/reptiles. Overlap in species detection between the two methods for mammals and birds was assessed using Venn diagrams created using the R package VennDiagram. Given that camera trapping is primarily effective for detecting mammals and birds, the comparison was restricted to these two taxonomic groups.

The costs of the two methods were calculated using Microsoft Excel. For camera trapping, costs included the purchase and maintenance of cameras, batteries, and SD cards. For eDNA, costs encompassed sampling kits, water quality testing instruments (e.g., pH meters), DNA extraction, PCR amplification, and sequencing expenses. The lifespan of the cameras was estimated based on observed losses in the study area, which included malfunctions, vandalism, and other factors. Since fieldwork for both camera trapping and eDNA sampling was conducted simultaneously, transportation and field labor costs were excluded from the analysis. Additionally, no extra costs were considered for species identification from camera trap images or bioinformatics analysis of eDNA data. Two cost comparisons were performed: (1) total cost for each method under sufficient sampling (species accumulation curve slope = 0.1) and (2) the average cost per species detected at each sampling point for eDNA or per camera for camera trapping.

To assess the relative importance of each eDNA sampling site, we defined an index that incorporated four key metrics: site-specific endemic species, alpha diversity, contribution to total beta diversity, and species rarity. Site-specific endemic species refers to the number of species detected at a given site that were recorded only once across the entire survey. Alpha diversity (species richness) represents the species richness of each sampling site. Contribution to total beta diversity was quantified as the absolute change in overall β -diversity (calculated using Sørensen’s index) following the removal of a given sampling site (Legendre and De Cáceres, 2013). Species rarity was defined as the inverse of the frequency with which a species was detected across all sampling sites. All four metrics were normalized for each site and then summed with equal weights to calculate a final relative importance score for each sampling site. We further explored the relationship between the relative importance of each sampling site and environmental variables, including

stream flow, pH, electrical conductivity (EC), temperature, rainfall on the day of sampling, and rainfall the day prior to sampling. This relationship was modeled using generalized additive modeling (GAM) and stepwise regression based on the Akaike Information Criterion (AIC). We also compared environmental variables across sampling sites between the two seasons to identify key factors that contributed to seasonal differences in the relative importance of sampling sites.

To estimate the transport distance of terrestrial vertebrate eDNA in rivers, we examined the altitudinal distributions of species detected in different seasons by both methods using the R package ggplot2. To ensure sufficient data for analysis while accurately reflecting seasonal changes in species' elevation distributions, we defined the low-rainfall season as September to November 2023 and the high-rainfall season as February to April 2024 for the camera trapping data. For the eDNA data, we only included results from tributaries, as the trunk stream does not accurately reflect the vertical distribution of species by elevation. We applied a GAM to examine the relationship between the elevation difference and the actual distance between two eDNA sampling sites. The "actual distance" was defined as the distance between two sampling points within the same watershed, connected by the physical water network, and was calculated using the Network Analysis tool of QGIS 3.34.10 (QGIS Development Team, 2024). This relationship was then used to infer the transport distance of eDNA within the river system.

Building on this, we used species detection data from both camera traps and eDNA to fit occupancy models for nine species using the PGOcc function from the R package spOccupancy, with data from each method analyzed separately (Doser et al., 2022). We generated response curves to illustrate the relationship between occupancy probability and elevation for each species. To better visualize the elevation distribution of the corresponding eDNA and camera trapping sampling sites, rug plots were incorporated into the figure. To more accurately capture patterns in species-specific elevation distributions, Bayesian priors of varying strengths were applied during model fitting, based on the characteristics of the detection data.

We calculated pairwise β -diversity for eDNA and camera trapping data separately using the R package betapart (Baselga and Orme, 2012). This included total β -diversity (β SOR, Sørensen dissimilarity), the turnover component (β SIM), and the nestedness component (β SNE; Baselga and Orme, 2012). To compare β -diversity patterns with the distance between sampling points for both methods, we plotted scatterplots fitted with locally estimated scatterplot smoothing (LOESS) curves. Only birds and mammals were included for this analysis as species' activity capacities and ecological habits vary across taxa and may influence eDNA transport. Additionally, limiting the analysis to birds and mammals allowed for more meaningful comparisons with camera trapping data. Since species detected by camera trapping tend to homogenize over time, we segmented the camera trapping data into three time periods for both seasons: one month, three months, and six months, each of which included the eDNA sampling period. The distance between eDNA sampling sites was calculated as the distance between two sites within the same watershed, connected by a physical water network, using QGIS 3.34.10. For camera trapping, the distance between two camera traps was calculated as the Least-Cost Path (LCP), which considers the complex topography of mountainous terrain and better reflects actual animal movement (Etherington and Penelope Holland, 2013). The LCP was computed using the costDistance function in the R package gdistance (van Etten, 2017). Additionally, eDNA sampling sites where the number of detected species was less than or equal to one were excluded from the β -diversity analysis, as these sites would prevent proper calculation of β -diversity metrics.

3. Results

3.1. Camera trapping

A total of 32,226 camera trap days were accumulated, yielding valid data from 143 cameras between April 30, 2023, and April 30, 2024, with 19,200 photos assigned to 2444 independent events. Of these events, 1926 (78.80 %) were identified as involving wildlife species (Figure S1), 337 (13.79 %) involved humans and domesticated animals, and 181 (7.40 %) comprised unidentified birds or rodents that could not be classified to species level. The domesticated animals observed included dogs (*Canis lupus familiaris*), cattle (*Bos taurus*), and horses (*Equus ferus caballus*). A total of seven wild bird species were detected, all from the Phasianidae family. These birds are key protected species in China, including Sclater's monal (*Lophophorus sclateri*), listed as endangered (EN) on China's Red List. We identified 32 wild mammal species, representing six orders and 15 families. These were predominantly medium- and large-sized species from the orders Cetartiodactyla, Primates, and Carnivora. Among these mammals, 20 species (62.50 %) are classified as threatened (including vulnerable [VU], endangered [EN], and critically endangered [CR]), and 24 species (75.00 %) are protected.

Notably, 53.85 % of species were detected in fewer than 10 independent events, whereas only 12.82 % were detected in 100 or more independent events. The most frequently observed bird was the blood pheasant (*Ithaginis cruentus*; NO = 0.098), while the most common mammal was the Gongshan muntjac (*Muntiacus gongshanensis*; NO = 0.406). Other frequently and widely observed species included the takin (*Budorcas taxicolor*), white-cheeked macaque (*Macaca leucogenys*), rhesus monkey (*Macaca mulatta*), and leopard cat (*Prionailurus bengalensis*). The yellow-bellied weasel (*Mustela kathiah*), Asiatic golden cat (*Catopuma temminckii*), leopard (*Panthera pardus*), and Phayre's leaf-monkey (*Trachypithecus melamera*) were each observed only once at a single site. The cumulative camera trap curve indicates that the sampling effort was relatively sufficient (Figure S2).

3.2. eDNA metabarcoding

Six of the 47 eDNA sampling sites did not yield replicates due to inaccessibility during the high-rainfall season. The remaining sites were sampled in both seasons at the same locations, resulting in a total of 264 replicates. Additionally, 16 negative controls were collected; however, 3 of these were discarded due to failure to support follow-up experiments.

During the low-rainfall season, eDNA metabarcoding identified 229 OTUs, assigned to 51 terrestrial vertebrate species and 6 fish

species. The terrestrial vertebrates included 7 domestic species and 44 wild species, which were further classified into 34 mammals, 7 birds, and 3 amphibians. Unidentified OTUs were classified at the family level, including Muridae, Soricidae, and Bovidae. Of the wild terrestrial vertebrates identified, 20.45 % were protected species, and 40.91 % were classified as threatened according to the Chinese Red List. The elegant water shrew (*Nectogale elegans*) was the most frequently detected (NO = 78.72 %), while 81.82 % of species were observed at no more than five sites. Species accumulation curves for all taxa, except birds, plateaued (Fig. 2a).

In the high-rainfall season, eDNA metabarcoding identified 526 OTUs, which were assigned to 69 terrestrial vertebrate species and 1 fish species. The terrestrial vertebrates included 7 domestic species and 62 wild species, which were classified into 36 mammals, 18 birds, 6 amphibians, and 1 reptile (Figure S1). Among the terrestrial vertebrates, 26.23 % were protected species, and 24.59 % were listed as threatened on the Chinese Red List. The elegant water shrew had the highest relative abundance (85.36 %), and 54.10 % of species were detected at no more than five sampling sites. Species accumulation curves for all taxa leveled off (Fig. 2b).

Regarding species composition, 15 species (19.74 %) were detected exclusively during the low-rainfall season, 32 species (42.10 %) were detected exclusively during the high-rainfall season, and 29 species (38.16 %) were detected in both seasons. As for sampling sites, most species (approximately 75 % in the low-rainfall season and 82 % in the high-rainfall season) were detected at the first sampling site, located before each first-order tributary merged with the mainstem. This suggests that biodiversity information is transported downstream by the river to a certain extent. On average, approximately 10 species were detected per sampling site during the high-rainfall season, compared to an average of approximately 4 species per sampling site during the low-rainfall season.

3.3. Comparison of eDNA and camera trapping detection effectiveness and detection costs

A total of 99 terrestrial vertebrate species were detected using both eDNA and camera trapping, including 64 mammals (64.65 %), 27 birds (27.27 %), 1 reptile (1.01 %), and 7 amphibians (7.07 %) (Figure S1). Of these, 16 species were detected by both methods, including 13 mammals (e.g., dhole *Cuon alpinus*) and 3 birds (e.g., blood pheasant *Ithaginis cruentus*; Fig. 2), with similar detection overlap observed across both seasons. The two methods exhibited complementary detection patterns. Camera trapping primarily identified large, ground-dwelling species from the orders Cetartiodactyla, Carnivora, Primates, and Galliformes. In contrast, eDNA metabarcoding detected a broader range of taxa but showed limited sensitivity for species with low occupancy probability as recorded by camera trapping (e.g., *Panthera pardus* and *Neofelis nebulosa*).

To assess the relative cost-effectiveness of camera trapping and eDNA metabarcoding for biodiversity monitoring, we calculated the

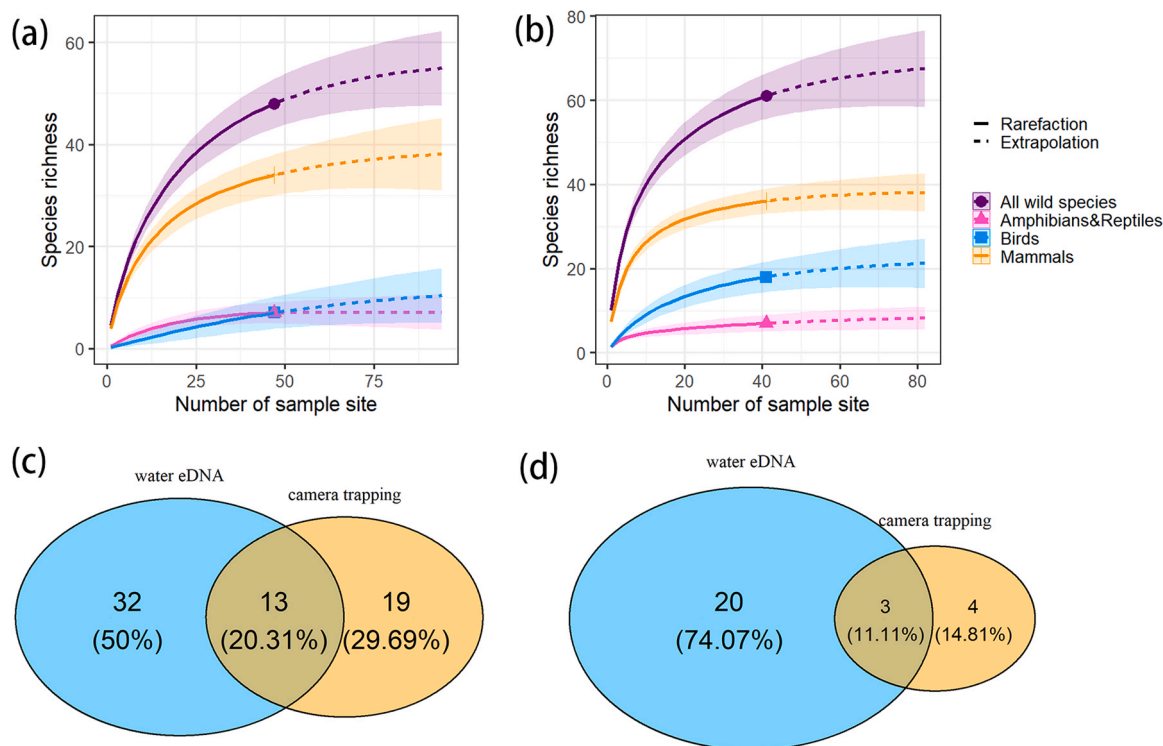


Fig. 2. Species accumulation curves based on eDNA data for both seasons and Venn diagrams comparing mammal and bird species identified using both methods. (a) Species accumulation curves for all terrestrial vertebrates and individual taxonomic groups based on low-rainfall season eDNA samples. (b) Species accumulation curves for all terrestrial vertebrates and individual taxonomic groups based on high-rainfall season eDNA samples. (c) Venn diagrams comparing mammal species detected through eDNA metabarcoding and camera trapping. (d) Venn diagrams comparing bird species detected through eDNA metabarcoding and camera trapping.

survey costs from field sampling to final species list (Table S3&S4). For camera trapping, 119 and 105 cameras were required to achieve a species accumulation curve slope of 0.1 during the May–October 2023 and November 2023–April 2024 monitoring cycles, respectively, at costs of \$31,403.00 and \$3221.00. The second monitoring cycle's costs primarily reflected camera replenishment, SD card replacement, and battery renewal. For eDNA metabarcoding, 86 and 77 sample points were needed to achieve the same slope for both seasons, with associated costs of \$7686.00 and \$6473.00, respectively.

Our results indicate that while the initial deployment of camera trapping incurs higher costs, subsequent maintenance costs are significantly lower. In contrast, eDNA metabarcoding exhibited smaller variation in cost per survey. The cost per species detected by camera trapping was \$28.00 and \$19.00 for the two monitoring cycles, whereas the cost per species detected by eDNA metabarcoding was \$23.00 and \$11.00 for the respective seasons.

3.4. eDNA sample site relative importance

In the low-rainfall season, eDNA sampling sites located in the mainstem and first-order tributaries exhibited significantly higher relative importance (Kruskal–Wallis test, $p = 0.024$; Fig. 3). However, some second-order tributary sites, such as GZ02 and LWD02, demonstrated high importance. In the high-rainfall season, although visual inspection suggested higher relative importance at sampling points in first- and second-order tributaries, the difference was not statistically significant (Kruskal–Wallis test, $p = 0.133$). This was due to the highest relative importance being observed at the mainstem site QLD01. Among the four standardized metrics, species rarity in both seasons showed significant correlations (Spearman's rank correlation test) with both site-specific endemic species and alpha diversity.

The most parsimonious generalized additive model (GAM) based on stepwise regression revealed seasonal differences in site relative importance. In the low-rainfall season, the only predictors with significant effects were water temperature (Tem, $p < 0.01$) and electrical conductivity (EC), with temperature having a stronger effect. In the high-rainfall season, the model identified three significant predictors: temperature (Tem, $p < 0.01$), electrical conductivity (EC, $p < 0.001$), and rainfall on the day before sampling (PCP_BS, $p < 0.05$). Water temperature and conductivity were significantly higher in the low-rainfall season compared to the high-rainfall season (Tem: paired t -test, $p < 0.001$; EC: Wilcoxon signed-rank test, $p < 0.001$). Additionally, significant seasonal differences were observed at the same eDNA sampling sites in pH (paired t -test, $p < 0.001$), rainfall on the sampling day (Wilcoxon signed-rank test, $p < 0.001$), and rainfall on the day before sampling (Wilcoxon signed-rank test, $p < 0.001$). However, flow rate showed no significant difference between seasons (Wilcoxon signed-rank test, $p = 0.417$).

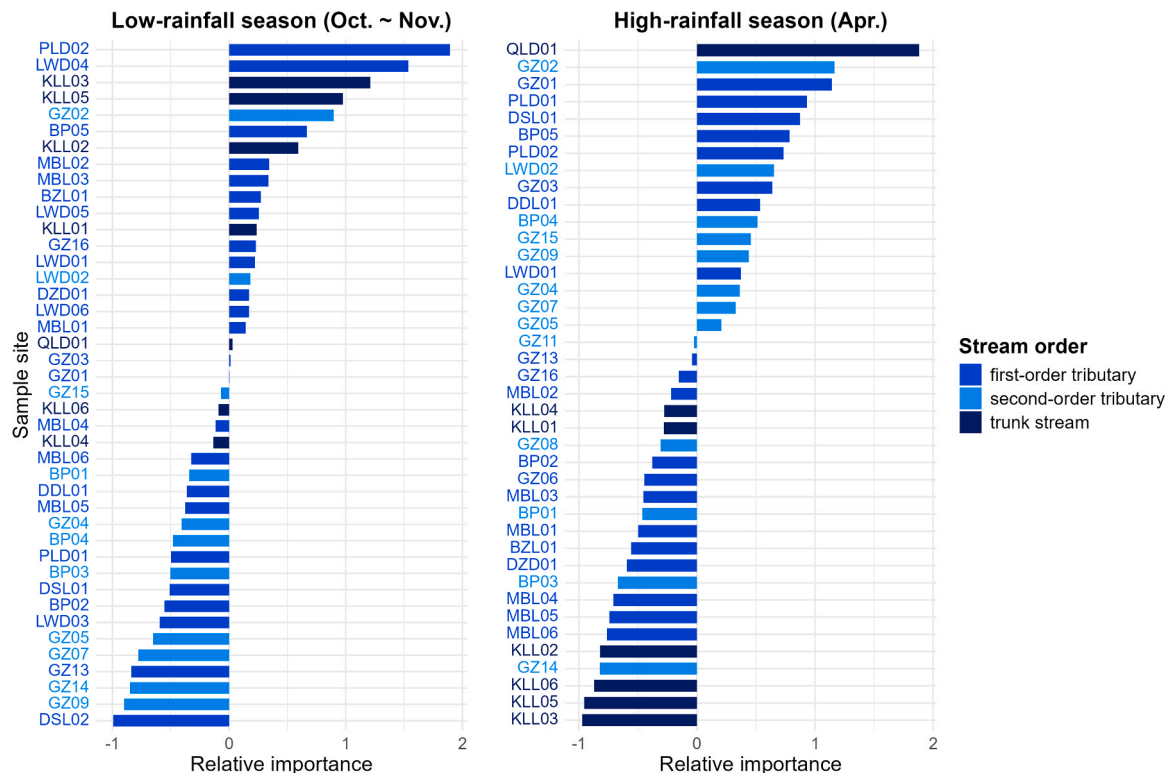


Fig. 3. Ranking of relative importance for each eDNA sampling site during both seasons. Relative importance was calculated based on four metrics: endemic species richness at each site, alpha diversity, contribution to overall beta diversity, and species rarity index.

3.5. eDNA transport distance of terrestrial vertebrate

We found that eDNA predominantly detected species at lower elevations, while camera trapping primarily recorded them at higher elevations (Figures S3&4). The response curves of occupancy probabilities for nine species, including the takin, along elevation gradients also revealed consistent differences between the two detection methods (Fig. 4). The mean elevation difference between species detected by the two methods was 748.98 m in the low-rainfall season and 784.79 m in the high-rainfall season, although the altitudinal distribution of camera trapping sites was significantly higher than that of eDNA sampling sites ($p < 0.001$, 95 % CI: 242.58–664.67 m), we conclude that camera trapping reflects the true altitudinal ranges of species in this region, given the concentration of human settlements and significant anthropogenic disturbance at lower elevations (Hu et al., 2025). The detection of eDNA from these species at lower elevations, where their presence is less likely, indicates potential upstream-to-downstream transport of eDNA.). To explore this further, we constructed a generalized additive model (GAM) to analyze elevation differences between sampling points in relation to riverine transport distance (Figure S5b). Using the mean elevation differences of 749.0 m and 784.8 m, we estimated that eDNA from species detected by both methods was transported approximately 14,104.8 m (95 % CI: 11,430.44–16,

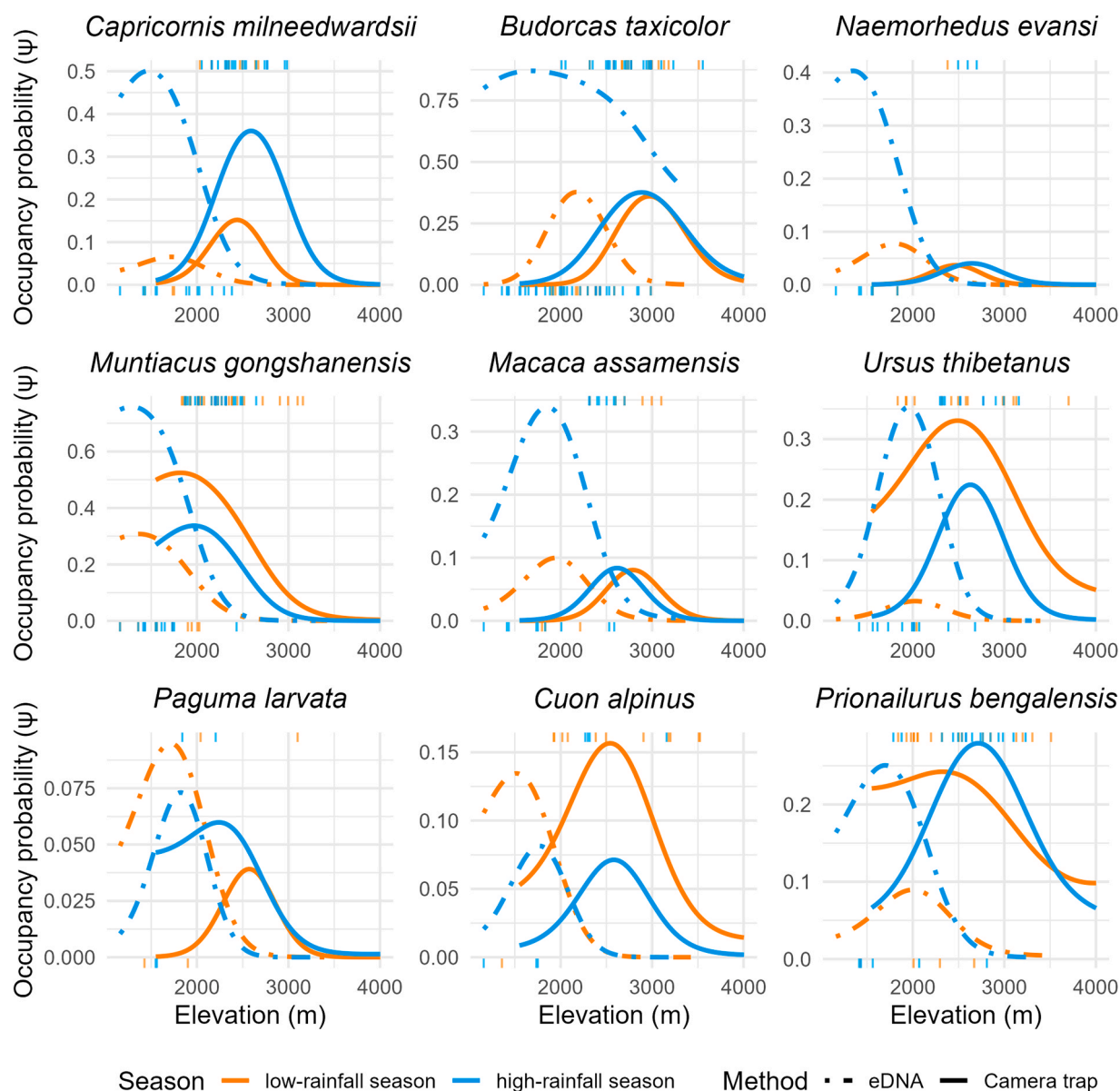


Fig. 4. Response curves of occupancy probability to elevation for nine species. Solid lines represent camera trap records, and dashed lines represent water eDNA records; orange indicates the low-rainfall season, and blue indicates the high-rainfall season. Rug plots below and above the X-axis indicate the elevation of water eDNA sampling sites and camera trapping sites, respectively.

779.1 m) during the low-rainfall season, and 15,074.2 m (95 % CI: 12,266.7–17,881.8 m) during the high-rainfall season.

For species detected by the same method in both seasons, there were some differences in elevation distribution. Specifically, larger animals detected by camera trapping (including ungulates, carnivores, and pheasants) were primarily distributed at lower elevations between February and April. During the low-rainfall season, eDNA tended to detect larger animals at lower elevation sites, whereas in the high-rainfall season, smaller mammals and birds were more commonly detected by eDNA at lower elevation sample sites.

We found that species composition similarity among camera traps increased with longer detection periods, as indicated by a decrease in initial pairwise β -diversity (Fig. 5). In the low-rainfall season, changes in pairwise β -diversity were not as pronounced due to low species detection rates at eDNA sampling sites. Therefore, we present only the high-rainfall season eDNA results here (Fig. 5), with low-rainfall season results provided in the [Supplementary Materials](#) (Figure S5). During the high-rainfall season, both pairwise β -diversity and its turnover component for eDNA increased significantly with greater distances between sampling sites. Notably, the initial pairwise β -diversity values for eDNA were lower than those derived from camera trapping, suggesting that eDNA detected at two sampling sites exhibited greater species composition similarity than the actual community composition. As the distance between sampling sites increased, the pairwise β -diversity values for eDNA converged with those from camera trapping at distances of 10–15 km.

4. Discussion

Our study demonstrates that the overlap in species detection between eDNA and camera trapping is primarily driven by species with larger body sizes and wider distribution (Figures 2 and S1), consistent with our predictions. Larger body sizes enhance detectability by camera traps, while more widespread distributions and larger population sizes increase the presence of their DNA in the environment (Lyet et al., 2021). This aligns with previous findings in temperate wetlands (Holm et al., 2023), tropical rainforests (Sales et al., 2020a), and temperate uplands (Sales et al., 2020b), where the two methods complement each other in detecting terrestrial vertebrates. Our results further emphasize that integrating multiple detection methods improves the overall detection of terrestrial vertebrate communities across diverse climatic zones. In particular, eDNA metabarcoding detected a broader range of taxa compared to camera trapping, especially for small-bodied species (Figure S1), which underscores a significant advantage of eDNA (Broadhurst et al., 2021; Leempoel et al., 2020). However, in other studies, eDNA did not consistently outperform camera trapping, likely because the latter also targeted aquatic and semi-aquatic species (Holm et al., 2023).

Consistent with previous studies (Broadhurst et al., 2021; Lyet et al., 2021), our results show that eDNA is less effective at detecting large felids, such as leopards and clouded leopards (Figure S1). In addition to their low population densities in the study area, their weak associations with aquatic habitats may also contribute to the low detection rates (Reji Chacko et al., 2023). In contrast, the elegant water shrew was detected at most eDNA sampling sites during both seasons, with detection rates of 78.72 % in the low-rainfall season and 85.36 % in the high-rainfall season, which can be explained by its semi-aquatic adaptations (Smith et al., 2008). Interestingly, eDNA also detected most flying squirrel species present in the region. As a strictly arboreal group, these species have generally been considered unlikely to be detected through aquatic eDNA surveys (Mena et al., 2021; Figure S1). While the exact mechanism for this detection is unclear and separate from the issue of downstream transport, it highlights the potential for aquatic eDNA to capture unexpected components of the terrestrial community. Given the limited understanding of their population status, ecological behavior, and life history traits (Li et al., 2021), the mechanisms underlying this unexpected result merit further investigation. Furthermore, only one reptile species, the Doria's ground skink (*Scincella doriae*), was detected across all eDNA samples, suggesting that terrestrial reptiles may not be ideal candidates for detection through water-derived eDNA compared to other taxa (Nordstrom et al., 2022; Reji Chacko

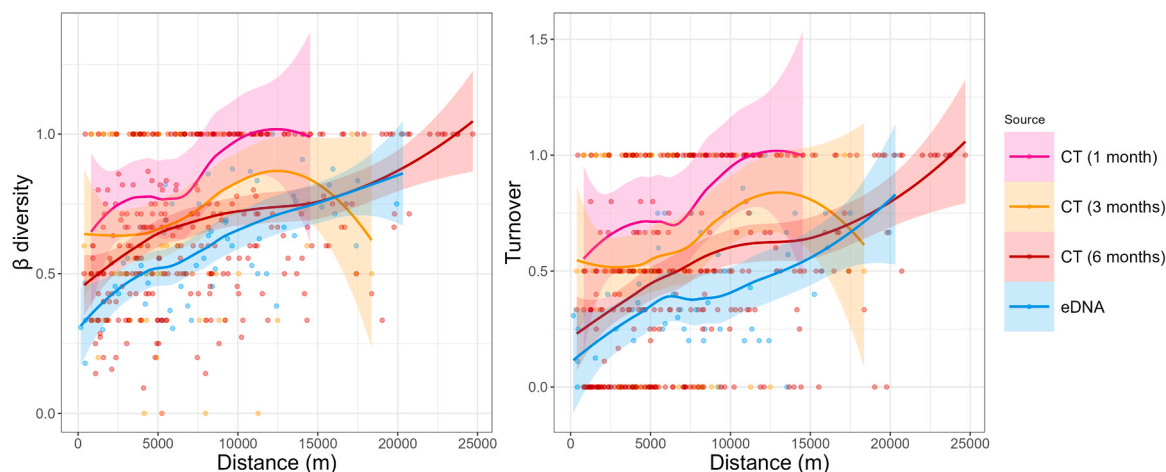


Fig. 5. Patterns of pairwise beta diversity with distance recorded by eDNA and three time periods of camera trapping during the rainy season. Both β -diversity (Sørensen dissimilarity) and its turnover component are displayed, with data sources differentiated by color. LOESS smoothing curves and 95 % confidence intervals are included to illustrate trends.

et al., 2023). Notably, the absence of rhesus monkeys and white-cheeked macaques (both in the genus *Macaca*) in the eDNA results may reflect the insufficient taxonomic resolution of the barcodes. While eDNA has been shown to be more cost-effective than camera trapping for surveying terrestrial mammals (Lyet et al., 2021; Mena et al., 2021), our study not only extends the survey to encompass the entire terrestrial vertebrate community, but also highlights that eDNA during the high-rainfall season provides valuable insights into terrestrial vertebrate diversity at a relatively low cost (Table S1).

More than half of the species identified through eDNA were detected in only one season (Figure S1). We propose two possible explanations for this pattern. First, the species detected in different seasons were predominantly from the orders Rodentia and Galiliformes, and their seasonal activity patterns may account for their detection in distinct seasons (De Souza et al., 2016). Second, varying seasonal climatic conditions, particularly rainfall, could influence species detection, with more species being detected during the high-rainfall season (Lyet et al., 2021). Although there were exceptions, the relative importance of the sampling sites showed distinct patterns between the two seasons (Fig. 3). For example, during the low-rainfall season, sites GZ02 and LWD02 located in second-order tributaries exhibited higher relative importance. This may be due to their location within the two largest second-order tributaries in the watershed area (94.38 km² and 29.45 km², respectively). Conversely, during the high-rainfall season, site QLD01 located in the trunk stream had the highest relative importance. This might be likely because it is situated farther from other sampling sites—at least 6 km away in a straight line—and represents a distinct vertebrate community.

Although the simplest GAM for the relative importance of sampling sites during the low-rainfall season did not include rainfall as an explanatory variable, we propose that rainfall plays a key role in shaping the relative importance of sampling sites across seasons. During the high-rainfall season, rainfall reduces both electrical conductivity (EC) and temperature in streams, with this effect likely being more pronounced in smaller streams. Rainfall also transports animal DNA (e.g., feces) from inland areas into streams and may facilitate the wet scavenging of airborne particles, enriching eDNA inputs into the stream system (Lynggaard et al., 2024). Additionally, spring snowmelt at higher elevations during the high-rainfall season may have increased the likelihood of detecting species at high-elevation sampling sites.

Our results contribute to the ongoing debate between ‘spatial design’ and ‘catchment design’ (Broadhurst et al., 2021; Lyet et al., 2021; Reji Chacko et al., 2023). While sampling during dry or low-rainfall seasons in trunk streams or first-order tributaries can reliably capture local terrestrial vertebrate diversity, we recommend expanding the sampling approach to include different stream classes during high-rainfall season to better represent true biodiversity levels.

The transport distance of eDNA released by aquatic organisms, such as fish, can be modeled by integrating factors like the particle size distribution of eDNA, its settling rate, and temperature-driven degradation (Pont, 2024). In contrast, the ecological processes governing eDNA from terrestrial animals—from production to detection in water—are more complex and not yet fully understood. It remains unclear whether DNA from terrestrial species enters water directly through behaviors like drinking or is indirectly transported by wind or other organisms. To address this, we developed a novel method for quantifying eDNA transport distances from terrestrial animals in rivers (Fig. 4&5). This approach, which utilizes species distribution data from camera traps, allows for the estimation of eDNA transport distances for entire communities, rather than focusing on single species, by analyzing changes in species composition across sampling points.

Our results indicate that the eDNA of terrestrial vertebrates in mountain rivers is not confined to specific elevation gradients (Reji Chacko et al., 2023), nor does it consistently accumulate at downstream sampling sites (Lyet et al., 2021). This contrasts with our expectation that sample sites would capture relatively localized diversity. The observed transport distances (10–15 km) are similar to those reported for aquatic eDNA (Deiner and Altermatt, 2014), though other studies have found considerably different transport distances (Jane et al., 2015; Pont et al., 2018). River flow is likely a key factor influencing eDNA transport downstream. Lower flow rates may reduce the efficiency of sediment and eDNA movement (Reji Chacko et al., 2023). For example, Jane et al. (2015) reported a downstream transport distance of 239.5 m for fish DNA at flow rates below 10 L/s at most sites. Pont et al. (2018) suggested that sampling intervals should be adjusted according to river size to minimize noise from transported eDNA and ensure sample independence. Additional factors, such as the volume of filtered water, sampling time, and river pH, may also affect the detectability of upstream species at downstream sites (Lyet et al., 2021; Perry et al., 2024). These variables highlight the complexity of interpreting eDNA transport dynamics in river systems.

In addition to generating species inventories, biodiversity monitoring efforts also aim to assess species abundance and distribution patterns. For aquatic eDNA, however, inferring the true spatial distribution of species remains challenging due to the downstream transport of DNA with water flow. Based on the results of this study, aquatic eDNA appears limited in its ability to resolve species distributions at fine spatial scales. Nevertheless, we argue that at broader landscape scales, the capacity of aquatic eDNA to integrate signals from multiple upstream sources allows it to effectively capture regional biodiversity patterns. This integrative property makes it a highly efficient, repeatable, and cost-effective approach for large-scale biodiversity monitoring.

In conclusion, our study underscores the cost-effectiveness of eDNA compared to camera trapping for short-term terrestrial vertebrate monitoring and demonstrates that the taxa detected by both methods are largely complementary. Our findings further highlight that the ability of eDNA collected from the same sites to detect terrestrial vertebrate diversity differs across seasons. This underscores the importance of adopting season-specific sampling protocols in future studies to more comprehensively capture species diversity. We inferred that bird and mammal eDNA in subtropical mountain rivers is transported 10–15 km downstream, based on camera trapping records of actual species distributions and by integrating altitudinal distribution patterns and pairwise β -diversity changes with distance between sampling points. These findings offer valuable insights for advancing future research on terrestrial vertebrate eDNA.

5. CRediT authorship contribution statement

Hongjiao Wang: Project administration. **Kenneth Otieno Onditi:** Writing – review & editing. **Shuiwang He:** Investigation. **Zhechang Hu:** Investigation. **Wenqiang Hu:** Investigation, Data curation. **Yinqiu Ji:** Writing – review & editing, Validation, Supervision, Software, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Ru Bai:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xueyou Li:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Xuelong Jiang:** Supervision, Funding acquisition. **Jianglin Zhang:** Investigation. **Yun Xiong:** Investigation.

Ethics Statement

Not applicable: This manuscript does not include human or animal research.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.gecco.2025.e03854](https://doi.org/10.1016/j.gecco.2025.e03854).

Data availability

I have shared link to my data at the Attach File step

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